28.3: SAR is a highly diverse group of protists defined by DNA similarities

A phylogenetic tree highlights the SAR group.

Our second supergroup, referred to as **SAR**, was proposed recently based on whole-genome DNA sequence analyses. These studies found that three major clades of protists—the stramenopiles, alveolates, and rhizarians—form a monophyletic supergroup. This supergroup contains a large, extremely diverse collection of protists. To date, this supergroup has not received a formal name but is instead known by the first letters of its major clades: SAR.

Some morphological and DNA sequence data suggest that two of these groups, the stramenopiles and alveolates, originated more than a billion years ago, when a common ancestor of these two clades engulfed a single-celled, photosynthetic red alga. Because red algae are thought to have originated by primary endosymbiosis (see Figure 28.3), such an origin for the stramenopiles and alveolates is referred to as secondary endosymbiosis. Others question this idea, noting that some species in these groups lack plastids or their remnants (including any trace of plastid genes in their nuclear DNA).

Despite its lack of a formal name, the SAR supergroup represents the best current hypothesis for the phylogeny of the three large protist clades to which we now turn.

Stramenopiles

One major subgroup of SAR, the **stramenopiles**, includes some of the most important photosynthetic organisms on the planet. Their name (from the Latin *stramen*, straw, and *pilos*, hair) refers to their characteristic flagellum, which has numerous fine, hairlike projections. In most stramenopiles, this “hairy” flagellum is paired with a shorter “smooth” (nonhairy) flagellum **(Figure 28.11)**. Here we’ll focus on three groups of stramenopiles: diatoms, oomycetes, and brown algae.

Figure 28.11

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Stramenopile flagella.
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Stramenopile have hairy and smooth flagella. Scale: 5 micrometers.

Most stramenopiles, such as Synura petersenii, have two flagella: one covered with fine, stiff hairs and a shorter one that is smooth.

Diatoms

A key group of photosynthetic protists, **diatoms** are unicellular algae that have a unique glass-like wall made of silicon dioxide embedded in an organic matrix **(Figure 28.12)**. The wall consists of two parts that overlap like a shoe box and its lid. These walls provide effective protection from the crushing jaws of predators: Live diatoms can withstand pressures as great as 1.4 million kg/m2, equal to the pressure under each leg of a table supporting an elephant!

Figure 28.12

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The diatom *Triceratium morlandii.*
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Diatom pictured is roughly triangular in shape with rounded “points” of the triangle and has a tough-looking outer edge. Scale: 40 micrometers.

(colorized SEM)

With an estimated 100,000 living species, diatoms are a highly diverse group of protists (see Figure 28.5). They are among the most abundant photosynthetic organisms both in the ocean and in lakes: One bucket of water scooped from the surface of the sea may contain millions of these microscopic algae. The abundance of diatoms in the past is also evident in the fossil record, where massive accumulations of fossilized diatom walls are major constituents of sediments known as *diatomaceous earth*. These sediments are mined for their quality as a filtering medium and for many other uses.

Diatoms are so widespread and abundant that their photosynthetic activity affects global carbon dioxide (CO_2) levels. Diatoms have this effect in part because of events that occur during episodes of rapid population growth, or *blooms*, when ample nutrients are available. Typically, diatoms are eaten by a variety of protists and invertebrates, but during a bloom, many escape this fate. When these uneaten diatoms die, their bodies sink to the ocean floor. It takes decades, or even centuries, for dead diatoms that sink to the ocean floor to be broken down by bacteria and other decomposers. As a result, the carbon in their bodies remains there for some time, rather than being released immediately as CO_2 as the decomposers respire. The overall effect of these events is that CO_2 absorbed by diatoms during photosynthesis is transported, or “pumped,” to the ocean floor.

With an eye toward reducing global warming by lowering atmospheric CO_2 levels, some scientists advocate promoting diatom blooms by fertilizing the ocean with essential nutrients such as iron. In one study testing this idea, researchers found that CO_2 was indeed pumped to the ocean floor after iron was added to a small region of the ocean, causing a diatom bloom. Other scientists have cautioned that the small area and short

duration of such studies makes it difficult to predict what would happen if large-scale iron fertilization experiments were performed. Indeed, the possibility that iron fertilization could have undesirable side effects (such as oxygen depletion or the production of nitrous oxide, a more potent greenhouse gas than CO_2) led the 2008 United Nations Convention on Biological Diversity to issue a moratorium against such experiments.

Brown Algae

The largest and most complex algae are **brown algae**. All are multicellular, and most are marine. Brown algae are especially common along temperate coasts that have cold-water currents. They owe their characteristic brown or olive color to the carotenoids in their plastids.

Many of the species commonly called “seaweeds” are brown algae. Some brown algal seaweeds have specialized structures that resemble organs in plants, such as a rootlike **holdfast**, which anchors the alga, and a stemlike **stipe**, which supports the leaflike **blades(Figure 28.13)**. Unlike plants, however, brown algae lack true tissues and organs. Moreover, morphological and DNA data show that these similarities evolved independently in the algal and plant lineages and are thus analogous, not homologous. In addition, while plants have adaptations (such as rigid stems) that provide support against gravity, brown algae have adaptations that enable their main photosynthetic surfaces (the leaflike blades) to be near the water surface. Some brown algae accomplish this task with gas-filled, bubble-shaped floats. Giant brown algae known as kelps that live in deep waters have such floats in their blades, which are attached to stipes that can rise as much as 60 m from the seafloor—more than half the length of a football field.

Figure 28.13

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Seaweeds: adapted to life at the ocean’s margins.
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Sea palm attached to rocky bottom by holdfast (resembles a branched plant root). Stipe is stalk-like coming out of holdfast and blades, at top, are flat and leaf-like in appearance.

The sea palm (Postelsia) lives on rocks along the coast of the northwestern United States and western Canada. The body of this brown alga is well adapted to maintaining a firm foothold despite the crashing surf.

Brown algae are important commodities for humans. Some species are eaten, such as *Laminaria* (Japanese “kombu”), which is used in soups. In addition, the cell walls of brown algae contain a gel-forming substance, called algin, which is used to thicken many processed foods, including pudding and salad dressing.

Alternation of Generations

A variety of life cycles have evolved among the multicellular algae. The most complex life cycles include an ****alternation of generations**, the alternation of multicellular haploid and diploid forms, as shown in **Figure 28.14** for the brown alga *Laminaria*. Although haploid and diploid conditions alternate in *all* sexual life cycles—human gametes, for example, are haploid—the term *alternation of generations* applies only to life cycles in which both haploid and diploid stages are multicellular. As you will read in Concept 29.1, alternation of generations also evolved in plants.

Figure 28.14

The life cycle of the brown alga *Laminaria:***an example of alternation of generations.**

Diagram outlines alternation of generations in Laminaria, from the diploid sporophyte through the haploid gametophyte generation and back to the sporophyte.

1. The sporophytes (2n) are usually found in water just below the line of the lowest tides, attached to the rocks by branching holdfast, with stipe and blades extending up towards the surface.
2. Cells on the surface of the blade develop into sporangia.
3. Sporangia produce haploid zoospores with both hairy and smooth flagella by meiosis.
4. The zoospores are all structurally alike, but about half of them develop into male gametophytes and half into female gametophytes. (Both are haploid). The gametophytes are short, branched filaments that grow on subtidal rocks.
5. Male gametophytes release haploid sperm, and female gametophytes produce haploid eggs, which remain attached to the female gametophyte. Eggs secrete a chemical signal that attracts sperm of the same species, thereby increasing the probability of fertilization in the ocean.
6. Sperm fertilize the eggs on the female gametophyte, resulting in diploid zygotes still on the haploid mature female gametophytes.
7. The zygotes grow into new diploid multicellular sporophytes while attached to the remains of the haploid female gametophyte. The diploid sporophyte continues to develop until the cycle begins again (#1). Photograph of a cluster of sporophyte blades is shown at scale 10 centimeters.

VISUAL SKILLS *Based on this diagram, are the sperm shown in* **STEP 5** *genetically identical to one another? Explain.*

*For suggested answer, see**Appendix A*.

As detailed in Figure 28.14, the diploid individual in this complex algal life cycle is called the *sporophyte* because it produces spores. The spores are haploid and move by means of flagella; they are called zoospores. The zoospores divide by mitosis and develop into haploid, multicellular male and female *gametophytes*, which produce gametes. The union of two gametes (fertilization) results in a diploid zygote, which matures and gives rise to a new multicellular sporophyte.

In *Laminaria*, the two generations are **heteromorphic**, meaning that the sporophytes and gametophytes are structurally different. Other algal life cycles have an alternation of **isomorphic** generations, in which the sporophytes and gametophytes look similar to each other, although they differ in chromosome number.

Oomycetes (Water Molds and Their Relatives)

Oomycetes include the water molds, the white rusts, and the downy mildews. Based on their morphology, these organisms were previously classified as fungi (in fact, oomycete means “egg fungus”). For example, many oomycetes have multinucleate filaments (hyphae) that resemble fungal hyphae **(Figure 28.15a)**. However, there are key differences between oomycetes and fungi. Among them, oomycetes typically have cell walls made of cellulose, whereas the walls of fungi consist mainly of another polysaccharide, chitin. Data from molecular systematics have confirmed that oomycetes are not closely related to fungi. In both oomycetes and fungi, the high surface-to-volume ratio of filamentous structures enhances the uptake of nutrients from the environment.

Figure 28.15

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Oomycetes.
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Two micrographs. Part a. Closeup of comycete hyphae. Part b. Water mold growing on a goldfish.

Although oomycetes descended from plastid-bearing ancestors, they no longer have plastids and do not perform photosynthesis. Instead, they typically acquire nutrients as decomposers or parasites. Most water molds are decomposers that grow as cottony masses on dead algae and animals, mainly in freshwater habitats **(Figure 28.15b)**. White rusts and downy mildews generally live on land as plant parasites.

The ecological impact of oomycetes can be significant. For example, the oomycete *Phytophthora infestans* causes potato late blight, which turns the stalks and stems of potato (and tomato) plants into black slime. Late blight contributed to the devastating Irish famine of the 19th century, in which a million people died and at least that many were forced to leave Ireland. The disease remains a major problem today, causing

crop losses as high as 70% in some regions. Globally, about $1 billion is spent each year on pesticides to control the parasite.

Alveolates

Members of the next subgroup of SAR, the **alveolates**, have membrane-enclosed sacs (alveoli) just under the plasma membrane **(Figure 28.16)**. Alveolates are abundant in many habitats and include a wide range of photosynthetic and heterotrophic protists. We'll discuss three alveolate clades here: a group of flagellates (the dinoflagellates), a group of parasites (the apicomplexans), and a group of protists that move using cilia (the ciliates).

Figure 28.16

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Alveoli.
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A micrograph and diagram show alveoli sacs adjacent to the flagellum.

These sacs under the plasma membrane are a characteristic that distinguishes alveolates from other eukaryotes (TEM).

Dinoflagellates

The cells of many **dinoflagellates** are reinforced by cellulose plates. Grooves in this "armor" house two flagella, one of which causes dinoflagellates (from the Greek *dinos*, whirling) to spin as they move through the waters of their marine and freshwater communities **(Figure 28.17a)**.

Figure 28.17

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Dinoflagellates.
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Two photos of a dinoflagellate and a red tide.

Parts are as follows.

Part a. Image: Spherical cell with groove in center (resembles a hamburger); spiral flagella inside groove. Scale: 3 micrometers. Dinoflagellate flagella. Beating of the spiral flagellum, which lies in a groove that encircles the cell, makes this specimen of Pfiesteria shumwayae spin (colorized SEM).

Part b. Image: Beach with red water lapping against sand and bluer water further out. Red tide in the Gulf of Carpentaria in northern Australia. The red color is due to high concentrations of a carotenoid-containing dinoflagellate.

Video

Dinoflagellate

image

Play

image

Play

Although their ancestors may have originated by secondary endosymbiosis (see Figure 28.3), roughly half of all dinoflagellate species are now purely heterotrophic. Others are important species of *phytoplankton* (photosynthetic plankton, which include photosynthetic bacteria as well as algae); many photosynthetic dinoflagellates are mixotrophic.

Periods of explosive population growth (blooms) in dinoflagellates sometimes cause a phenomenon called "red tide" **(Figure 28.17b)**. The blooms make coastal waters appear brownish red or pink because of the presence of carotenoids, the most common pigments in dinoflagellate plastids. When blooms occur, toxins produced by certain dinoflagellates have caused massive kills of invertebrates and fishes. Humans who eat molluscs that have accumulated the toxins are affected as well, sometimes fatally. Based on data from 1982–2016, researchers at Stony Brook University found that ocean warming (caused by ongoing climate change) has expanded the potential for algal blooms in two species of toxic dinoflagellates, making them a growing threat to human health.

Apicomplexans

Nearly all **apicomplexans** are parasites of animals—and virtually all animal species examined so far are attacked by these parasites. The parasites spread through their host as tiny infectious cells called *sporozoites*. Apicomplexans are so named because one end (the *apex*) of the sporozoite cell contains a *complex* of organelles specialized for penetrating host cells and tissues. Although apicomplexans are not photosynthetic, recent data show that they retain a modified plastid (apicoplast), most likely of red algal origin.

Most apicomplexans have intricate life cycles with both sexual and asexual stages. Those life cycles often require two or more host species for completion. For example, *Plasmodium*, the parasite that causes malaria, lives in both mosquitoes and humans **(Figure 28.18)**.

Figure 28.18

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    The two-host life cycle of *Plasmodium*, the apicomplexan
that causes malaria.
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Person is bitten by infected Anopheles mosquito; Plasmodium sporozoites enter the liver cells, and then merozoites infect red blood cells. Gametocytes are produced and lifecycle continues in another mosquito host.

Diagram depicts Plasmodium life cycle in both the human and the mosquito host.

1.An infected Anopheles mosquito bites a person, injecting Plasmodium sporozoites (haploid) in its saliva.

2.The sporozoites enter the person's liver cells. After several days, the sporozoites undergo multiple divisions and become merozoites (haploid), which use their apical complex to penetrate red blood cells. Diagram shows numerous merozoites spilling out of a liver cell and penetrating red blood cells. TEM shows a merozoite on the surface of a red blood cell attaching at its apex. Scale: 0.5 micrometers.

3.The merozoites divide asexually inside the red blood cells. At intervals of 48 or 72 hours (depending on the species), large numbers of merozoites break out of the blood cells, causing periodic chills and fever. Some of the merozoites infect other red blood cells.

4.Some merozoites form gametocytes (haploid).

5.Another Anopheles mosquito bites the infected person and picks up Plasmodium gametocytes along with blood.

6.Gametes (haploid) form from gametocytes; each male gametocyte produces several slender male gametes. Female gametes are also formed.

7.Fertilization occurs in the mosquito's digestive tract, and a diploid zygote forms.

8.Following meiosis, an oocyst (haploid) develops from the zygote in the wall of the mosquito's gut. The oocyst releases thousands of haploid sporozoites, which migrate to the mosquito's salivary gland. Cycle can repeat at step (1).

Are morphological differences between sporozoites, merozoites, and gametocytes caused by different genomes or by differences in gene expression? Explain.

*For suggested answer, see**Appendix A.*

Animation

Life Cycle of a Malaria Parasite

image

image

Historically, malaria has rivaled tuberculosis as the leading cause of human death by infectious disease. The incidence of malaria was diminished in the 1960s by insecticides that reduced carrier populations of *Anopheles* mosquitoes and by drugs that killed *Plasmodium* in humans. But the emergence of resistant varieties of both *Anopheles* and *Plasmodium* has led to a resurgence of malaria. According to the World Health Organization, there were an estimated 241 million malaria cases and 627,000 malaria deaths worldwide in 2020. In regions where malaria is common, the lethal effects of this disease have resulted in the evolution of high frequencies of the sickle-cell allele; for an explanation of this connection, see Figure 23.18.

The search for malarial vaccines has been hampered by the fact that *Plasmodium* lives mainly inside cells, hidden from the host's immune system. And, like trypanosomes, *Plasmodium* continually changes its surface proteins. Even so, significant progress was made when European regulators recently approved the world's first licensed malarial vaccine (RTS,S or Mosquirix). By 2019, selected regions of Africa began routine use of this vaccine, and in 2021 the World Health Organization recommended it for broad use in children in areas where malaria is prevalent. However, this vaccine, which targets a protein on the surface of sporozoites, provides only partial protection against malaria. As a result, researchers continue to study other potential vaccine targets, including the apicoplast. This approach may be effective because the apicoplast is a modified plastid; as such, it descended from a cyanobacterium and hence has different metabolic pathways from those in human patients.

Ciliates

The **ciliates** are a large and varied group of protists named for their use of cilia to move and feed **(Figure 28.19a)**. Most ciliates are predators, typically of bacteria or of other protists. Their cilia may completely cover the cell surface or may be clustered in a few rows or tufts. In certain species, rows of tightly packed cilia function collectively in locomotion. Other ciliates scurry about on leg-like structures constructed from many cilia bonded together.

Figure 28.19

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Structure and function in the ciliate *Paramecium caudatum.*
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Two diagrams. Part a. Feeding, waste removal and water balance. Part b. Conjugation and reproduction.

Parts are as follows.

Part a. Paramecium constantly takes in water by osmosis from its hypotonic environment. Bladderlike contractile vacuoles accumulate excess water from radial canals and periodically expel it through the plasma membrane. Thousands of cilia cover the surface of the Paramecium. Cilia along a funnel-shaped oral groove move food (mainly bacteria) into the cell mouth, where the food is engulfed into food vacuoles by phagocytosis. Food vacuoles fuse with lysosomes (not shown). As the food is digested, the vacuoles follow a looping path through the cell. Wastes are released when the vacuoles fuse with a specialized region of the plasma membrane that functions as an anal pore. In the image, micronucleus are adjacent to and smaller than macronucleus. Contractile vacuoles (quantity 2) are distinguishable as star-like structures; food vacuoles as spheres; cilia as short “hairs” on cell surface. Scale: 50 micrometers.

Part b. Conjugation (Steps 1-4):

1. Two cells of compatible mating strains align side by side and partially fuse. The micronuclei are diploid.
2. Following meiosis, four haploid micronuclei are produced in each cell.
3. Three micronuclei in each cell disintegrate. The remaining micronucleus in each cell divides by mitosis.
4. The cells swap one micronucleus.

Asexual reproduction (Steps 5-9):

5. The cells separate.
6. Following micronuclear fusion, the two haploid micronuclei in the cell fuse into a diploid micronucleus.
7. Three rounds of mitosis produce 8 micronuclei inside the cell.
8. Four micronuclei become macronuclei and the original macronucleus disintegrates.
9. Two rounds of binary fission yield four daughter cells. Cells that are compatible mates can now continue at step 1.

MAKE CONNECTIONS *The events shown in steps* **4** *through* **6** *of this diagram have a similar overall effect to what event in the human life cycle (see*Figure 13.5*)*? Explain.*

*For suggested answer, see**Appendix A*.

Video

Paramecium

image

Play

image

Play

A distinctive feature of ciliates is the presence of two types of nuclei: tiny micronuclei and large macronuclei. A cell has one or more nuclei of each type. Genetic variation results from ****__conjugation__, a sexual process in which two individuals exchange haploid micronuclei but do not reproduce **(Figure 28.19b)**. Ciliates generally reproduce asexually by binary fission, during which the existing macronucleus disintegrates and a new one is formed from the cell's micronuclei. Each macronucleus typically contains multiple copies of the ciliate's genome. Genes in the macronucleus control the everyday functions of the cell, such as feeding, waste removal, and maintaining water balance.

Video

Ciliate Movement in *Stentor*

image

Play

image

Rhizarians

Our next subgroup of SAR is the **rhizarians**. Many species in this group are **amoebas**, protists that move and feed by means of ****__pseudopodia__, extensions that may bulge from almost anywhere on the cell surface. An amoeba moves by extending a pseudopodium and anchoring its tip; more cytoplasm then streams into the pseudopodium. Amoebas do not constitute a monophyletic group; instead, they are dispersed across many distantly related eukaryotic taxa. Most amoebas that are rhizarians differ morphologically from other amoebas by having threadlike pseudopodia. Rhizarians also include flagellated (non-amoeboid) protists that feed using threadlike pseudopodia.

We'll examine three groups of rhizarians here: radiolarians, forams, and cercozoans.

Radiolarians

The protists called **radiolarians** have delicate, intricately symmetrical internal skeletons that are generally made of silica. The pseudopodia of these mostly marine protists radiate from the central body **(Figure 28.20)** and are reinforced by bundles of microtubules. The microtubules are covered by a thin layer of cytoplasm, which engulfs smaller microorganisms that become attached to the pseudopodia. Cytoplasmic streaming then carries the captured prey into the main part of the cell. After radiolarians die, their skeletons settle to the seafloor, where they have accumulated as an ooze that is hundreds of meters thick in some locations.

Figure 28.20

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A radiolarian.
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Radiolarian. Scale: 200 micrometers.

Numerous threadlike pseudopodia radiate from the central body of this radiolarian (LM).

Forams

The protists called **foraminiferans** (from the Latin *foramen*, little hole, and *ferre*, to bear), or **forams**, are named for their porous shells, called **tests** (see Figure 28.5). Foram tests consist of a single piece of organic material that typically is hardened with calcium carbonate. The pseudopodia that extend through the pores function in swimming, test formation, and feeding. Many forams also derive nourishment from the photosynthesis of symbiotic algae that live within the tests.

Forams are found in both the ocean and fresh water. Most species live in sand or attach themselves to rocks or algae, but some live as plankton. The largest forams, though single-celled, have tests several centimeters in diameter.

Ninety percent of all identified species of forams are known from fossils. Along with the calcium-containing remains of other protists, the fossilized tests of forams are part of marine sediments, including sedimentary rocks that are now land formations. Foram fossils are excellent markers for correlating the ages of sedimentary rocks in different parts of the world. Researchers are also studying these fossils to obtain information about climate change and its effects on the oceans and their life **(Figure 28.21)**.

Figure 28.21

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Fossil forams.
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Fossil foram tests embedded in rock. Forams are roughly tear-shaped and lined with layers of lines.

By measuring the magnesium content in fossilized forams like these, researchers seek to learn how ocean temperatures have changed over time. Forams take up more magnesium in warmer water than in colder water.

Cercozoans

First identified in molecular phylogenies, the **cercozoans** are a large group of amoeboid and flagellated protists that feed using threadlike pseudopodia. Cercozoan protists are common inhabitants of marine, freshwater, and soil ecosystems.

Most cercozoans are heterotrophs. Many are parasites of plants, animals, or other protists; many others are predators. The predators include the most important consumers of bacteria in aquatic and soil ecosystems, along with species that eat other protists, fungi, and even small animals. One small group of cercozoans, the chlorarachniophytes (mentioned earlier in the discussion of secondary endosymbiosis), are mixotrophic: These organisms ingest smaller protists and bacteria as well as perform photosynthesis. At least one other cercozoan, *Paulinella chromatophora*, is an autotroph, deriving its energy from light and its carbon from CO_2. *Paulinella***(Figure 28.22)** appears to represent an intriguing additional evolutionary example of a eukaryotic lineage that obtained its photosynthetic apparatus directly from a cyanobacterium.

Figure 28.22

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A second case of primary endosymbiosis?
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Cercozoan Paulinella with chromatophore highlighted.

The cercozoan Paulinella conducts photosynthesis in a unique sausage-shaped structure called a chromatophore (LM). Chromatophore membranes include a peptidoglycan layer, indicating that they are derived from a bacterium. DNA evidence shows that chromatophores are derived from a different cyanobacterium than that from which plastids are derived.